



— **PORTFOLIO PROJECT 21 | 2026-2027**
Digital Twin-Inspired **Workflow Intelligence™**

A student-developed healthcare operations case study that turns an emerging digital-twin concept into a practical, no-PHI simulation of patient access, workflow readiness, exception ownership, scenario testing, interoperability, and responsible technology governance.

Created by Kori Pickle, BSHA Candidate, University of Phoenix. Educational, simulated, and no PHI. No real patient, employer, payer, claims, EHR, device, portal, or operational outcome data were used.

REVIEW SIMULATED CASE

[DOWNLOAD BRANDED PDF](#)

— **HEALTHCARE
OPERATIONS
INTELLIGENCE ENGINE™**

Model the workflow.
Test the pressure.
Protect the handoff.

CORE QUESTION

Where
did the
workflow
first lose
control?

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Candidate

The post is a **market signal**, not proof that healthcare has already mastered digital twins.

What it means

A digital twin aims to represent a real patient, organ, device, facility, or workflow so teams can monitor change, test scenarios, predict risk, and inform decisions.

Why it is growing

AI, machine learning, cloud computing, connected devices, simulation tools, and richer data environments make increasingly dynamic models possible.

Where value may appear

Personalized care, clinical research, equipment maintenance, bed and staffing planning, patient flow, and administrative workflow optimization.

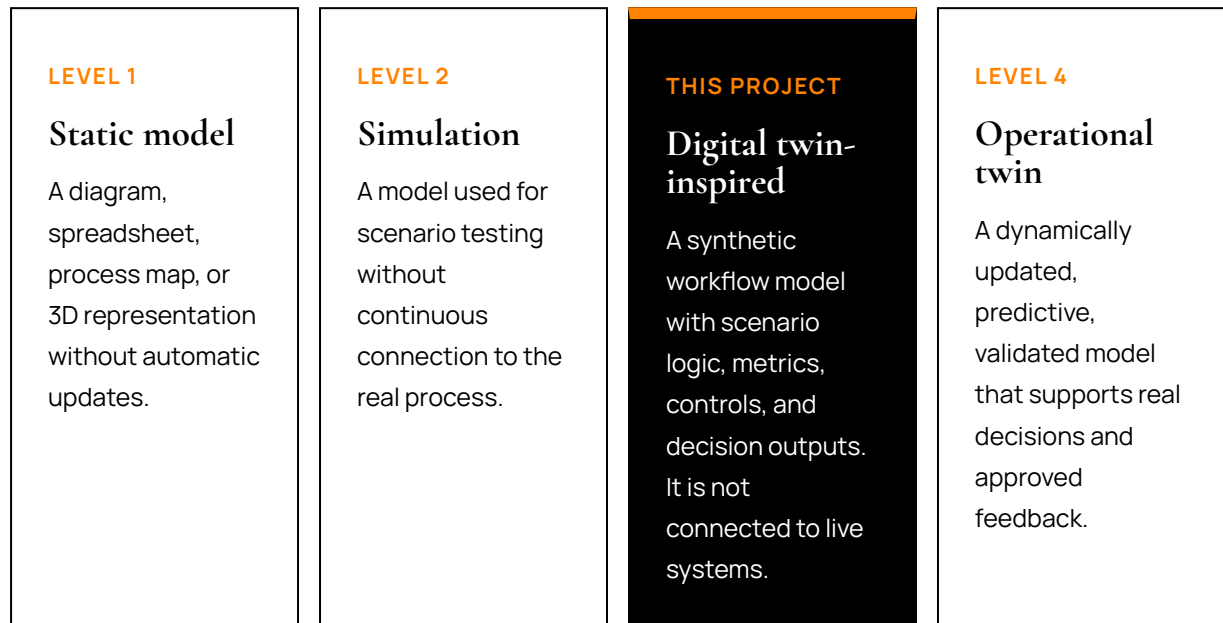
What the graphic cannot prove

A company logo wheel does not establish maturity, outcomes, market leadership, safety, equity, or interoperability. The companies shown operate at different technology layers.

The commercial forecast cited in the post is a market estimate, not a guarantee of adoption or benefit. Operational value still depends on data quality, validation, ownership, implementation, and patient safeguards.

Not every model is a **true digital twin**.

The National Academies defines a digital twin as a virtual construct that mimics a real system, is dynamically updated with data from that system, predicts behavior, and informs decisions. Bidirectional interaction is central.



Honest labeling is part of the career proof. This is a digital twin-inspired operational simulation, not a live clinical or health-system twin.

SIMULATED NO-PHI CASE

A specialty-care access workflow looks connected but keeps producing avoidable delay.

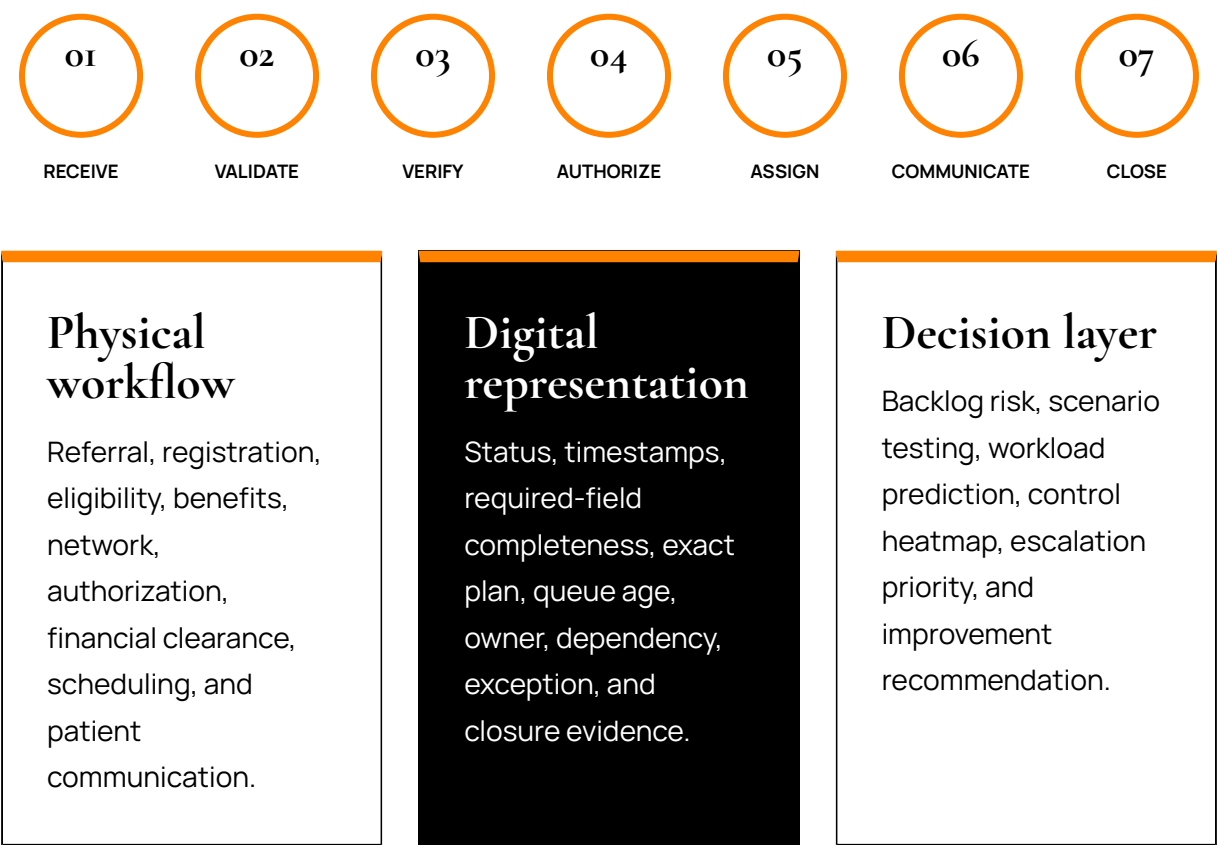
A simulated outpatient service receives 600 referrals per month across an EHR, payer portals, document queues, scheduling tools, and spreadsheets. No shared readiness model shows whether each case is complete, verified,

authorized, financially cleared, accepted by the next owner, and safe to schedule.

Referral intake	Incomplete referrals enter the queue without a visible acceptance decision.
Eligibility	Active coverage is sometimes mistaken for complete financial clearance.
Authorization	Requests age in separate payer workspaces, and missing information is found late.
Scheduling	Patients are contacted before readiness is confirmed, creating rescheduling and repeated calls.
Exception ownership	No single status shows who owns the next action or when escalation is required.
First loss of control	An incomplete referral is labeled ready enough to move forward. That false-ready status spreads uncertainty downstream.

The visible issue is scheduling delay. The upstream control failure is an unverified handoff from intake to readiness review.

Model the process so it can be **observed, tested, and improved.**



— **SYNTHETIC SCENARIO ANALYSIS**

Upstream controls may reduce downstream pressure.

All values are synthetic educational examples. They are not benchmarks, employer results, or real-world predictions.

Measure	Baseline	Standardized intake	Closed-loop readiness
First-pass completeness	71%	86%	92%
Referral-to-ready time	8.4 days	6.2 days	4.9 days

Measure	Baseline	Standardized intake	Closed-loop readiness
Cases aging beyond 7 days	22%	13%	7%
Avoidable rework per 100 referrals	31	18	9
Primary control	No shared readiness gate	Required-field checklist and rejection reason	Checklist, named owner, aging threshold, daily exception review, and verified closure

The strongest simulated improvement comes from defining what ready means, assigning the exception, and verifying closure - not merely adding more technology.

CAREER PROOF

What this project demonstrates without overstating experience.

Systems thinking

Traces a visible scheduling problem backward to a false-ready handoff.

Workflow analysis

Maps access, verification, authorization, scheduling, communication, and closure.

Data literacy

Defines synthetic fields, statuses, timestamps, aging measures, ownership, and evidence.

Scenario planning

Compares baseline and future-state controls using transparent assumptions.

Implementation readiness

Converts the concept into controls, exception paths, UAT thinking, and governance.

Patient-to-professional insight

Connects administrative uncertainty to repeated calls, rescheduling, delay, and confusion.

Resume project bullet

Developed a student-level, digital twin-inspired simulation of a specialty-care patient-access workflow using synthetic no-PHI data; mapped referral, eligibility, authorization, financial-clearance, scheduling, and exception controls; compared baseline and future-state scenarios; and documented data-quality, interoperability, governance, human-review, and patient-impact requirements.

PATIENT ACCESS

HEALTH INFORMATICS

WORKFLOW INTELLIGENCE

SCENARIO ANALYSIS

DATA QUALITY

INTEROPERABILITY

RESPONSIBLE AI

IMPLEMENTATION SUPPORT

EVIDENCE BASE AND DISCLAIMER

Selected references

1. National Academies of Sciences, Engineering, and Medicine. (2024). [Foundational Research Gaps and Future Directions for Digital Twins](#).
2. Katsoulakis, E., Wang, Q., Wu, H., et al. (2024). [Digital twins for health: A scoping review](#).
3. Ringeval, M., Etindele Soso, F. A., Cousineau, M., & Paré, G. (2025). [Advancing health care with digital twins](#).

4. Calcaterra, V., Guardamagna, L., Gatti, A., et al. (2026). [Digital twins in healthcare: A systematic review.](#)
5. Crawford, M. (2026). [How digital twins can improve health system operations.](#)

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STRATEGIC TAKEAWAY

Healthcare workflow intelligence starts by asking: Where did the workflow first lose control?